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Research Defence

I am Jannatul Ferdous-Nabila and today I am going to present my research titled:

Does a context file actually help? An Empirical, Task level comparison of Agent's Coding with and Without an Agent's Context File.

My research is based on the unidesk Full stack ticketing application, which uses FastAPI, React, PostgreSQL, SQL Alchemy and Alembic.

Research Motivation:

Nowadays, AI coding agents are becoming increasingly common in software development. Developers often provide these agents with a context file such as AGENT.md, which contains project rules, coding conventions and instructions.

However, I found an important question: Does an Agent's context file actually improve the coding agent's performance, or does it mainly control the agent's decisions?

Most claim about context files are based on developer experience rather than empirical comparison. So, I want to test this claim through an experiment.

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Research Questions:

My main research Question was:
Does a context file improve agent's coding performance, and if it does, where does the improvement actually happen?

My hypothesis was the AGENT.md would have stronger effect on large, ambiguous and high-risk task, while Vits effect would be negligible on small and well-defined tasks.

Methodology:

For the experiment, I selected five tasks with increasing levels of size and ambiguity.

The tasks included:

- i) Improve empty dashboard text.
- ii) Adding dashboard ticket search
- iii) Enforcing ticket lifecycle transitions.
- iv) Implementing ticket assignment and agent work queues.
- v) Hardening authentication and session management.

This task was performed under two conditions:

- i) Without AGENT.md
- ii) With AGENT.md

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Then I compared the result using four criteria: Correctness, convention, adherence, Unrequested extras, and edge-case handling.

Findings:

The most important finding was that small tasks showed almost no significant difference.

For example, in the first task, both versions produced essentially the same result. The task was so simple that there was very little opportunity for the agent to make a wrong decision.

But when the task became larger and more ambiguous, the difference became much more visible.

For example, in Task-4, without AGENT.md, the agent created an unapproved public endpoint and left an orphaned database migration after the code was reverted.

With AGENT.md, these problems were avoided because the context file required approval before changing public API paths and encouraged careful handling of migrations.

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Another important finding came from Task 5, authentication handling.

The version without AGENT.mn introduced two silent behavioral regressions that were not detected by the existing green test suite.

This showed me that passing tests does not always mean that the implementation is behaviorally correct.

Most Important Conclusion:

The most important conclusion of my research is:

AGENT.mn did not make the coding smarter. It made the agent make fewer unsupervised guesses.

In small task, there was almost no measurable benefit.

But in large, ambiguous, and high-risk tasks, the context file acted mainly as a governance mechanism.

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It restrict unnecessary decisions, prevent unapproved changes, and helped preserve important project rules.

However, the context file was not always better.

In one task, the AGENT.md-guided implementation become more conservative than the actual requirement. Also, the free-form version caught one security-related issue that the context-file version initially missed.

So, my research not conclude that the AGENT.md should always be used.

Insted, I conclude that it is most valuable for medium-to-large ambiguous, and high-risk tasks

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Limitations:

My research has several limitation.

First, I used only one repository and one technology stack.

Second, only five tasks were tested, so the sample size is small.

Third, the same agent was involved in producing and evaluating the result, so there is a possibility of evaluation bias.

Finally, I tested my one particular style of AGENT.md so the result may be different with another context file design.

Final statement:

To conclude, my research suggests that the value of a context file is not primarily about increasing the intelligence of an AI coding agent.

Instead, its main value is governance.

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It helps define what the agent should and should not change, when the task is complex and the cost of making a wrong assumption is high.

Therefore, my final keyword is,

The context file did not make the agent smarter — it made its unsupervised guess fewer.